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Urban Interceptor Drone Physics

Physics-grounded equations for modeling a counter-UAS interceptor drone in a built-up environment — kinematics, guidance, aerodynamics, energy, propulsion, sensing, and urban propagation.

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Urban Interceptor Drone Physics

A decision-support reference of closed-form, physics-grounded equations for modeling a counter-UAS interceptor drone in an urban environment. Each section gives the core relation, physical meaning, and an urban-specific note. Use these as deterministic building blocks for simulation, wargaming, what-if analysis, and synthetic scenario generation.

Boundary. This is for training / simulation / wargaming only. It is not weaponeering, not a live engagement model, and not an authorization to compute real-world lethality, collateral damage, or engagement permissibility. Keep simulation authority explicit and human-governed.

When to Use This Skill

Use when the user asks to model or reason about:

- Intercept geometry, time-to-intercept, or miss distance for a drone-on-drone engagement
- Proportional-navigation or pursuit guidance for a counter-UAS platform
- Turn radius / turn rate / load-factor limits in an urban canyon
- Energy maneuverability, climb/acceleration, or endurance of a small electric interceptor
- Radar or EO/IR detection of a low-RCS target with urban line-of-sight blocking
- Multipath, GPS-denied drift, or wind effects between buildings
- Scenario parameters or "what-if" numbers for any of the above

Conventions

- Bold symbols are vectors; plain symbols are scalars. SI units assumed.
- g = gravitational acceleration (≈ 9.81 m/s²), ρ = air density (≈ 1.225 kg/m³ at sea level, lower with altitude/temperature).
- Sub-sonic, low-Reynolds, rigid-body assumptions are used unless stated otherwise.

1. Kinematics

Position update under constant acceleration:

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0\, t + \tfrac{1}{2}\mathbf{a}\, t^2$$

Speed from initial conditions:

$$v(t) = v_0 + a\,t$$

Relative (target-to-interceptor) position and closing velocity:

$$\mathbf{R} = \mathbf{r}_T - \mathbf{r}_I, \qquad V_c = -\dot{R}, \qquad \dot{\mathbf{R}} = \mathbf{v}_T - \mathbf{v}_I$$

`Urban note:` closing velocity is high for head-on geometries, which shrinks time-to-intercept but also reduces reaction time; near-zero V_c (tail chase) degrades guidance observability.

2. Intercept Geometry

Time to intercept under constant closing rate (zero-effort miss basis):

$$t_f = \frac{R}{V_c}$$

Collision-triangle condition (both bodies on a constant course to a common point):

$$\frac{\mathbf{R}}{R} \approx \frac{\mathbf{v}_T - \mathbf{v}_I}{|\mathbf{v}_T - \mathbf{v}_I|}$$

Line-of-sight (LOS) angle and its rate:

$$\lambda = \operatorname{atan2}(R_y, R_x), \qquad \dot{\lambda} = \frac{(R_x \dot{R}_y - R_y \dot{R}_x)}{R^2}$$

Zero-effort miss distance (the miss that would occur with no corrective acceleration):

$$M_{ZEM} = R \cdot \dot{\lambda} \cdot t_f$$

3. Guidance

Classical proportional navigation (lateral acceleration command normal to LOS):

$$a_c = N\, V_c\, \dot{\lambda}$$

`N` = navigation gain, typically $N \in [3, 5]$; too low under-corrects, too high invites saturation and instability.

Augmented proportional navigation (command compensates target maneuver a_T):

$$a_c = N\, V_c\, \dot{\lambda} + \tfrac{N}{2}\, a_{T\perp}$$

Pure-pursuit (velocity vector drives onto LOS):

$$\dot{\psi} = k\, \varepsilon$$

with ε being the lead angle (heading error), k a pursuit gain.

Capture/terminal condition is met when miss distance $m(t_f) \le r_{kill}$, where r_{kill} is the effective intercept radius (for a kinetic/hard-kill interceptor, near-contact; for a non-kinetic or netting platform, a larger capture envelope).

`Urban note:` guidance gains that are stable in open air may ring or saturate near buildings because LOS is interrupted by occlusion; model LOS dropout as a $\dot{\lambda}$ freeze with an uncertainty growth term, not a clean continue.

4. Aerodynamics

Lift:

$$L = \tfrac{1}{2}\rho V^2 S C_L$$

Drag:

$$D = \tfrac{1}{2}\rho V^2 S C_D, \qquad C_D = C_{D0} + \frac{C_L^2}{\pi\, e\, AR}$$

Here S = reference area, C_L = lift coefficient, C_{D0} = parasitic drag coefficient, e = Oswald efficiency, AR = aspect ratio.

Load factor (normal acceleration in g):

$$n = \frac{L}{W}$$

Bank angle for a coordinated level turn:

$$\phi = \arccos\left(\frac{1}{n}\right)$$

Turn radius:

$$R_{turn} = \frac{V^2}{g\sqrt{n^2 - 1}}$$

Turn rate:

$$\dot{\psi} = \frac{g\sqrt{n^2 - 1}}{V}$$

`Urban note:` street widths (often 15–40 m) and building spacing set a hard geometric ceiling on usable turn radius; if $R_{turn} >$ corridor width, the interceptor cannot turn inside the canyon and must climb or accept a miss.

5. Energy Maneuverability

Specific energy (energy height):

$$E_s = h + \frac{V^2}{2g}$$

Specific excess power (instantaneous rate of energy change):

$$P_s = \frac{T - D}{W} V$$

Climb rate at a given condition:

$$\dot{h} = \frac{(T - D)\, V}{W}$$

Sustainable-load-factor constraint (max turn that does not bleed energy):

$$n_{ss} = \frac{L}{W} \quad \text{such that} \quad T = D$$

6. Propulsion & Endurance

Thrust-to-weight ratio for climb/acceleration capability:

$$TWR = \frac{T}{mg}$$

Power required for steady level flight:

$$P_{level} = D\, V = \frac{W}{L/D} V$$

Electric endurance (first-order):

$$t_{endur} = \frac{E_{bat}\, \eta_{system}}{P_{avg}}$$

Electric range (Breguet-with-battery analog):

$$R_{range} = \frac{\eta_{system}\, E_{bat}}{mg}\ \frac{L}{D}$$

`Urban note:` battery mass is the dominant design lever for small interceptors; endurance and maneuver (peak-power) demand trade against each other, so model peak-power draw during the intercept burst separately from loiter P_{avg}.

7. Sensing (Detection)

Monostatic radar range (single pulse, point target):

$$R_{max} = \left[\frac{P_t\, G^2\, \lambda^2\, \sigma}{(4\pi)^3\, k\, T_s\, B\, F\, (S/N)_{min}\, L}\right]^{1/4}$$

Terms: P_t transmit power, G antenna gain, λ wavelength, σ target radar cross-section (RCS), k Boltzmann constant, T_s system noise temperature, B bandwidth, F noise figure, $(S/N)_{min}$ minimum SNR, L system losses.

Signal-to-noise ratio at range R:

$$S/N = \frac{P_t\, G^2\, \lambda^2\, \sigma}{(4\pi)^3\, R^4\, k\, T_s\, B\, F\, L}$$

Detection probability under Swerling fluctuation models (Swerling 0–4) relates P_d to S/N and P_{fa}; for a small multirotor the RCS can be < 0.01 m², sharply limiting R_{max}.

EO/IR detection is range-limited by the diffraction/Nyquist resolution of the detector:

$$R_{acq} \approx \frac{d_{target}}{2\, \tan(\theta_{IFOV})}$$

with d_{target} the minimum resolvable target extent and θ_{IFOV} the instantaneous field of view per pixel.

`Urban note:` radar and RF links in the canyon suffer severe block-shadow and clutter; ground clutter RCS is high relative to a small drone, compressing the usable SNR.

8. Urban Occlusion & Propagation

Line-of-sight existence between sensor and target is a geometric block test (ray vs. building footprint at height), not a smooth function:

$$LOS_i(t) = \mathbf{1}\Big[\text{no building intersects segment } \overline{\mathbf{r}_S(t)\,\mathbf{r}_T(t)}\Big]$$

Fresnel-zone clearance for a working microwave link:

$$F_1 = 8.657\sqrt{\frac{d_1 d_2}{f\,(d_1+d_2)}} \quad (\text{m})$$

with d_1, d_2 distances (km) from obstacle to each end, f frequency (GHz). Signal degrades as the first Fresnel zone is encroached by rooftops.

Free-space path loss:

$$L_{fs} = \left(\frac{4\pi d}{\lambda}\right)^2$$

Multipath/urban excess loss is added empirically over free space; the combined received power:

$$P_r = P_t G_t G_r L_{fs}^{-1} L_{urban}^{-1} L_{pol}^{-1}$$

9. GPS-Denied Navigation

Dead-reckoning position error growth is governed by sensor quality and integration time. For velocity-noise-dominated error:

$$\sigma_{pos}(t) \approx \sigma_v\, t$$

For accelerometer-bias-dominated (double integration) error:

$$\sigma_{pos}(t) \approx \tfrac{1}{2}\,\delta_a\, t^2$$

where σ_v is velocity-measurement noise and δ_a is accelerometer bias. IMU orientation drift feeds these errors; urban GPS denial (multipath, shadows) forces reliance on VIO / lidar / optical flow with their own loop-closure limits.

`Urban note:` quantify drift as a growing uncertainty ball on the interceptor state; this drives terminal homing to a fused (not single-sensor) solution.

10. Wind & Gust

Urban canyon wind speed between buildings (Venturi acceleration — first-order):

$$U_{canyon} \approx U_{ambient} \frac{A_{upstream}}{A_{canyon}}$$

with effective cross-sectional areas upstream vs. in the street. Gust-induced aerodynamic force scales quadratically with relative airspeed:

$$F_{gust} \approx \tfrac{1}{2}\rho\,(V + u')^2 SC$$

`Uncertainty:` gust u' is stochastic; model as a bounded wind field, and add wind as an accelerating disturbance to the guidance loop rather than a static offset.

Quick Reference (Parameter Checklist)

- `N` (nav gain), `V_c`, $\dot{\lambda}$ → guidance command a_c
- T/W, C_L, C_D → turn / climb / energy bounds
- σ (RCS), P_t, B → radar range
- building footprints + heights → LOS mask & Fresnel clearance
- battery E_{bat}, η → endurance / intercept burst budget
- wind field → disturbance on guidance

Output Shape

When returning a what-if answer, state each formula used, the numeric inputs, the result with units, and (for urban cases) the dominant limiting term. Mark illustrative values as illustrative, and never present a computed miss distance or intercept as an authorization to engage.